



Diagram illustrating a quantum circuit element (a unitary gate) labeled $\hat{U}$, followed by an equivalence arrow pointing to its definition:

$$\hat{U} = \frac{1}{\sqrt{d}} \sum_{j,k=0}^{d-1} \omega_d^{jk} |j\rangle \langle k|, \quad \omega_d = e^{\frac{2\pi i}{d}}$$